

Student 13 **Jayaraman Renuka Devi** 192573007RD

5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 3.33 TC: 2/3 passed

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        s.rollNo=Integer.parseInt(sc.nextLine().trim());
        //sc.nextLine();
        s.name=sc.nextLine();
        s.marks=Double.parseDouble(sc.nextLine().trim());
        s.displayDetails();
    }
}
```

INPUT

```
101
Renuka Devi
89.5
```

OUTPUT

```
Roll No: 101
Name: Renuka Devi
Marks: 89.5
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Student{
    private int rollNo;
    private String name;
    private double marks;
    void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    void setName(String name){
        this.name=name;
    }
    void setMarks(double marks){
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);
        Student s= new Student();
        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());
        s.displayDetails();
    }
}
```

INPUT

```
101
renu
89
```

OUTPUT

```
Roll No: 101
Name: renu
Marks: 89.0
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }
    Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }
    void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
    }
}
```

INPUT

102
Renuka
90.5

OUTPUT

Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: Renuka
Marks: 90.5

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Salary{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empId=sc.nextInt();
        sc.nextLine();
        String empName=sc.nextLine();
        double basicSalary=sc.nextDouble();
        Employee e=new Employee();
        e.getEmployeeDetails(empId,empName,basicSalary);
        e.calculateSalary();
        e.displayDetails();
    }
}
class Employee{
    int empId;
    String empName;
    double basicSalary;
    double hra;
    double da;
    double netSalary;
    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId=empId;
        this.empName=empName;
        this.basicSalary=basicSalary;
    }
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        netSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Net Salary: " + netSalary);
    }
}
```

INPUT

```
101
renuka
60000
```

OUTPUT

```
Employee ID: 101
Employee Name: renuka
Basic Salary: 60000.0
HRA: 12000.0
DA: 6000.0
Net Salary: 78000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class GrossSalary{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empId=sc.nextInt();
        sc.nextLine();
        String empName=sc.nextLine();
        double basicSalary=sc.nextDouble();
        Employee e=new Employee();
        e.getEmployeeDetails(empId,empName);
        e.getBasicSalary(basicSalary);
        e.calculateSalary();
        e.displayDetails();
    }
}
class Employee{
    int empId;
    String empName;
    double basicSalary;
    double hra;
    double da;
    double grossSalary;
    void getEmployeeDetails(int empId,String empName){
        this.empId=empId;
        this.empName=empName;
    }
    void getBasicSalary(double basicSalary){
        this.basicSalary=basicSalary;
    }
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        grossSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Gross Salary: " + grossSalary);
    }
}
```

INPUT

```
101
Renuka
50000
```

OUTPUT

```
Employee ID: 101
Employee Name: Renuka
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Gross Salary: 65000.0
```